

Introduction & Objectives

The Problem: Modern geodetic data (GPS) is crucial for monitoring megathrust earthquakes but is temporally limited, with most records only beginning in the 1990s.

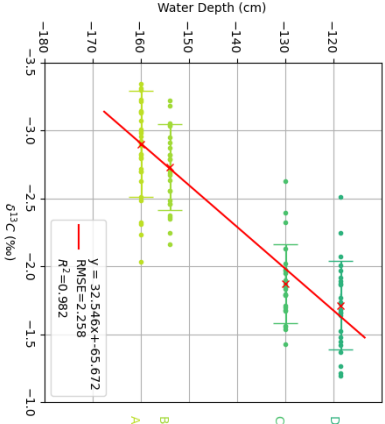
The Solution: We utilize coral skeletons as temporally high-resolved and purely vertical historical geodetic recorders in subduction settings.

Goal: To establish an annual-scale geodetic record that extends centuries back, providing a deeper understanding of the long-term earthquake cycle.

Geological Setting & Methodology

Solomon Islands

Study focused on the New Georgia Group, where the active trench is exceptionally close to the coastline (<30 km). This proximity makes the region highly sensitive to vertical deformation from subduction zone processes.



Methods

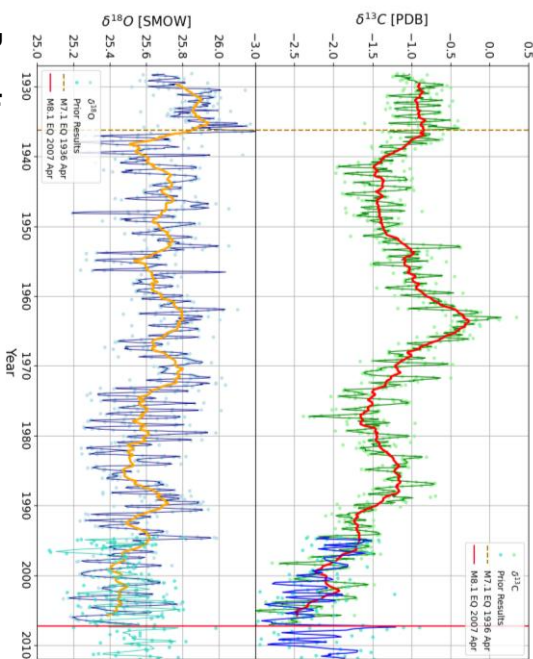
Sampling: Collected **575** horizontal and **141** vertical coral samples. We measured stable isotopes $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ to estimate relative sea level (RSL).

Calibration: Established a strong linear relationship between water depth and $\delta^{13}\text{C}$ by analyzing vertical samples from synchronized ages.

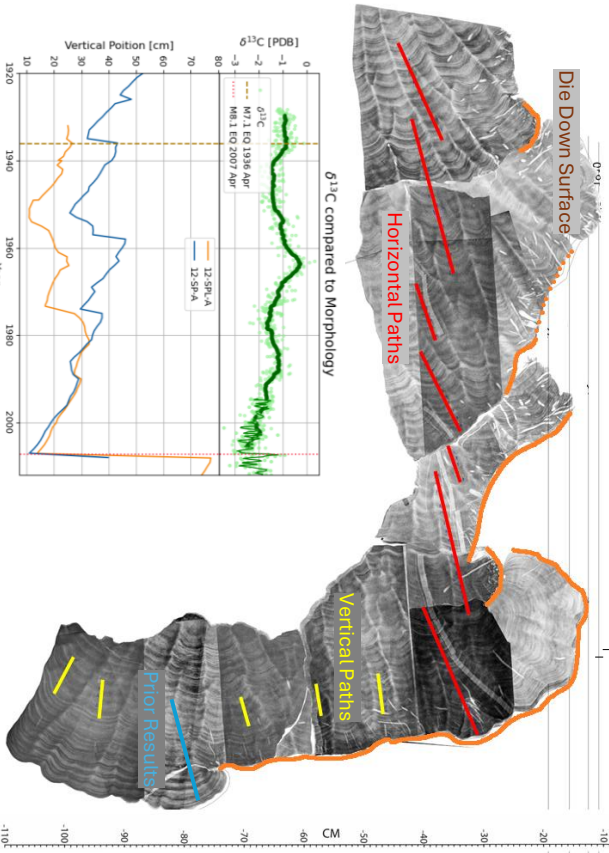
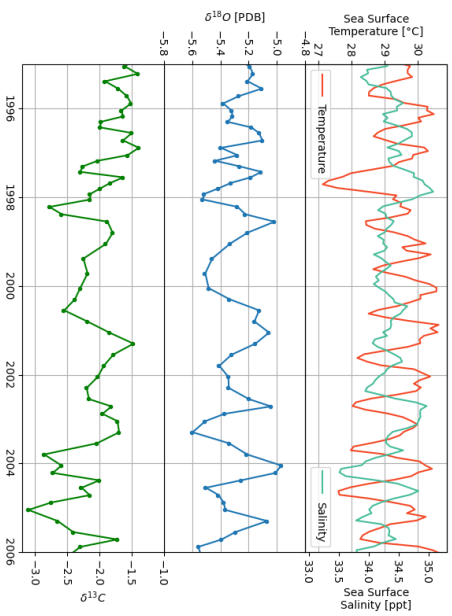


High-resolution Coral Geodesy in the Solomon Islands

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Results
Stable isotopes clearly record major regional seismic events:
• **1936 Event 7.0 MW:** Captured primarily in the $\delta^{18}\text{O}$
• **2007 Event 8.1 MW:** Strongly reflected in $\delta^{13}\text{C}$

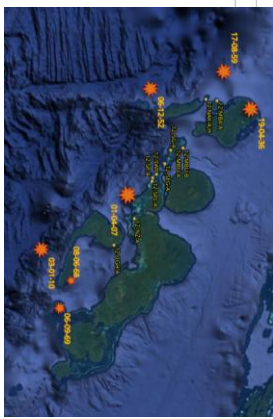
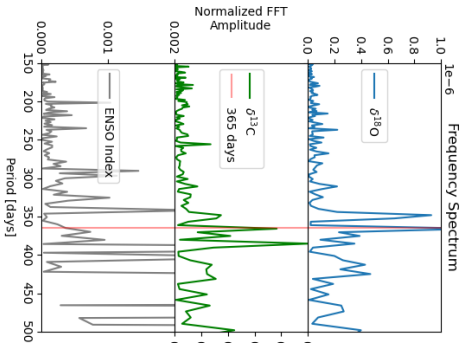


Decoupling Tectonic vs. Climatic Signals

- Isotopic signals are influenced by any local and global sea level change, Sea Surface Temperature (SST), Sea Surface Salinity (SSS), ENSO variability or Pacific Decadal Oscillation (PDO).
- By comparing coral data with external instrumental datasets, we can successfully isolated the **tectonic signal** from climatic noise. Also spectral tools help to filter high frequency parameters compared to tectonics.

Morphology vs. Geochemistry

- While coral morphology indicates RSL change, it is often "binary". It only records events that reach the low-tide level. **Isotope analysis** provides a continuous vertical record, even when the coral remains submerged.



Summary & Conclusions

- $\delta^{13}\text{C}$ correlates strongly with RSL and aligns with external environmental datasets (SST, SSS, and Tide Gauges).
- Coral morphology provides a structural "sanity check" that validates the isotopic trends.
- Major Finding:** We demonstrate that $\delta^{13}\text{C}$ is a robust indicator of vertical tectonics and serves as a powerful proxy for **paleogeodesy** in subduction environments.

Ongoing & Future Work

- Implementing a **bayesian age model** to refine the temporal resolution of the proxy.
- Scaling:** Expanding the analysis to a broader suite of coral samples for a comprehensive regional reconstruction of the Solomon Islands' seismic history.